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RESEARCH ARTICLE

BEKMP: A Blockchain-Enabled Key Management Protocol for Underwater Acoustic Sensor Networks

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ABSTRACT This paper introduces a new blockchain-assisted key management protocol specifically designed for clustered Underwater Acoustic Sensor Networks (UASNs). The protocol focuses on simplifying cross-cluster reauthentication for mobile underwater devices and reducing the risks of internal attacks caused by compromised nodes. To address the resource limitations of UASN devices, the protocol utilizes Elliptic Curve Qu Vanstone (ECQV) certificates, which considerably reduce Public Key Infrastructure (PKI) overhead. Moreover, it integrates the Hashed One-pass Menezes-QuVanstone (HOMQV) protocol with blockchain technology to enhance key exchange security against active attacks. The blockchain is hosted on resource-intensive surface nodes that serve as cluster heads and decentralized authorities for certificate management. Smart contracts embedded within the blockchain facilitate certificate-related operations and node trust scoring mechanism. We compared the proposed solution against a state-of-the-art benchmark and demonstrated that it imposes lower computational and communication overheads while exhibiting robustness against various attacks, as confirmed by the AVISPA tool. The solution is implemented on actual underwater/surface devices and validated experimentally. By leveraging the immutable, traceable, and fault-tolerant properties of blockchain, it establishes a secure, scalable, and efficient communication framework for mission-critical UASN applications.

INDEX TERMS Authentication, blockchain, underwater sensor network.

I. INTRODUCTION

As the scope of Underwater Acoustic Sensor Networks (UASNs) widens to include not just scientific research but also industrial operations and national security initiatives, the imperatives of safeguarding these networks are increasingly being recognized. UASNs are comprised of cooperating sensor nodes, which can be either anchored or mobile, and operate in a challenging underwater environment that is often inaccessible and susceptible to various malicious attacks.

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Ensuring reliable and secure communication in such settings has thus emerged as a vital research focus [1], [2], [3], [4].

While various security solutions have been effective in fortifying terrestrial wireless networks, their direct application to UASNs presents many challenges. This is largely due to the unique difficulties posed by underwater acoustic channels, such as very small signal propagation speed (around 1500m/s), narrow bandwidth, and significant signal attenuation. Due to large communication delays, wormhole attacks can be performed relatively easily by using an out-of-band radio connection above the water surface [5]. Furthermore, the limited bandwidth poses substantial challenges to implementing effective security protocols. Standard

countermeasures against replay attacks, like packet time-stamping, fall short in UASNs due to constraints on packet header size and the need to limit communication overhead in low data-rate conditions. Moreover, timestamp-based strategies risk the unintended loss of legitimate packets due to the significant Packet Delivery Delays (PDD), which can extend up to several minutes [6]. Adding to this complexity is GPS unavailability for network synchronization, making the use of timestamps both challenging and costly. Also, noteworthy is the increased energy consumption in UASNs, often an order of magnitude higher than their terrestrial counterparts. This further amplifies the need for research aimed at developing comprehensive security frameworks that balance the trade-off between minimizing communication overhead and energy consumption with maintaining optimal security levels.

In large-scale UASN deployments, ensuring secure communication is a pressing challenge. Such networks typically employ a cluster-based structure to enhance energy efficiency and reduce delays [7]. Within this structure, a Cluster Head (CH) — typically the node with the least energy constraints— collects, processes, and forwards data of its affiliated nodes to the terrestrial network infrastructure. One crucial aspect of these networks is the frequent movement of nodes between clusters, whether intentional or unintentional. A central security concern in this scenario is establishing a robust device authentication framework. Considering the movement of sensor nodes, the cross-cluster authentication process should be both fast and efficient. While many authentication methods based on symmetric cryptography or digital signatures exist in the literature, most of them face issues such as a single point of failure or performance limitations due to their centralized architecture [8]. These limitations can result in considerable re-authentication latency as sensor nodes transition between network clusters.

To address the above-mentioned challenges, recent research advocates for decentralized authentication approaches [8], [9], [10]. Blockchain technology, wherein a distributed network of nodes maintains uniform data without a centralized authority, stands out as a viable solution. Through blockchain, trust can be established across diverse domains, with each domain having a dedicated node to maintain a shared global ledger [11]. Implementing blockchain not only safeguards against potential breaches targeting databases managed by a central entity but also reduces inter-domain communication overhead [12]. Motivated by these facts several blockchain-based authentication approaches have been proposed for Internet of Things (IoT) networks [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], including Internet of Underwater Things (IoUT) [8], [9]. These methods address authentication by means of asymmetric cryptography, mostly relying on certificates from trusted Certificate Authorities (CAs) to establish a verifiable association between an entity and its corresponding public key. Yet, these methods largely overlook certificate life-cycle management, which is

crucial for the overall security infrastructure, ensuring that compromised or outdated credentials do not undermine the network. Additionally, these methods often do not account for the limited resources of sensor nodes. Our paper aims to fill this gap by introducing a blockchain-based authentication and key management framework that facilitates cross-cluster re-authentication for mobile underwater devices. The system integrates implicit certificates to offer not only secure but also lightweight solution, well-suited for very resource-constrained environments. The blockchain exclusively resides on resource-intensive surface nodes that operate as CHs, providing a decentralized infrastructure for key and certificate management. Smart contracts, embedded within the blockchain, govern all certificate-related operations and further support a trust evaluation mechanism for sensor nodes, enabling automated certificate revocation. Our approach also incorporates the Hashed One-time Menezes-Qu-Vanstone (HOMQV) protocol [23] to facilitate efficient session key exchange in a single message, thereby conserving energy. Replay attack vulnerabilities inherent to this protocol are mitigated by recording communication logs on the blockchain, providing a secure, verifiable history that prevents malicious retransmissions.

The key contributions of the paper are summarized as follows:

- We present an effective blockchain-based authentication and key management system for UASNs. The proposed solution enables streamlining of cross-cluster reauthentication for underwater devices and mitigating the risks of internal attacks via compromised nodes.
- The proposed authentication method has been designed to be lightweight, fast and secure. To account for resource constraints of underwater sensor nodes, a layered network architecture is assumed, where blockchain is implemented only on resource-powerful surface nodes acting as CHs. The latency and energy consumption are optimized by employing implicit certificates and the HOMQV protocol for key exchange.
- We have conducted a thorough comparative analysis of the proposed authentication method against the existing state-of-the-art, examining security capabilities, communication and computation overheads, and energy efficiency. Additionally, the security of the proposed scheme has been validated using the AVISPA tool [24].
- The proposed security framework is implemented on authentic UASN devices and its performance in terms of authentication/re-authentication latency has been assessed through sea-trials.
- The complete code of the framework implementation and simulations is publicly available in our GitHub repository.¹

The structure of the paper is as follows: Section II provides a summary of related work in the field. The considered

¹<https://github.com/monusen-uom/BEKMP-Blockchain-Platform>.
<https://github.com/monusen-uom/Desert-BEKMP>

network model and the proposed blockchain-based key management protocol is detailed in Section III. Security evaluation of the proposed protocol is presented in Section IV. Section V analyzes communication and computational overheads, as well as energy efficiency. Finally, the paper concludes with Section VI.

II. RELATED WORK

UASNs are becoming increasingly important for various maritime applications, including, marine environment monitoring, target detection and tracking, commercial exploration, surveillance of critical infrastructure, and many others [2], [3]. The distinct nature of underwater acoustic channels, combined with the resource constraints of underwater sensor devices, makes UASNs particularly vulnerable to a wide array of malicious attacks. This vulnerability is underscored in several recent studies [2], [3], [25], [26], which not only highlight the unique characteristics of UASNs but also provide a comprehensive analysis of potential threats, attacks, and corresponding mitigation strategies. In this section, we provide a comprehensive overview of related work, organized into three key themes: i) Mitigating External Attacks, ii) Mitigating Internal Attacks, and iii) Blockchain-Assisted Security. Additionally, we highlight the distinctive contributions of our paper.

A. MITIGATING EXTERNAL ATTACKS

A range of cryptography-based solutions have been proposed in the literature for the security of UASNs. In [27], the authors proposed using Elliptic Curve Cryptography (ECC) on underwater acoustic modems. They used Digital Signal Processors (DSPs) to speed up cryptographic computations. By efficiently implementing ECC, they significantly reduced computational requirements. In [28], the suitability of different digital signature schemes for the authentication problem in UASNs, particularly in terms of energy consumption, was examined. The study from [25] presents the most effective cryptographic primitives that can be used to secure UASNs at various protocol layers ranging from the network layer to the application layer. The authors proposed a security framework that offers data confidentiality, integrity, authentication, and non-repudiation, and is built on Advanced Encryption Standard (AES) [29] in the Galois Counter Mode (GCM) and short digital signature algorithms. However, it is important to note that none of these proposed solutions directly address the challenge of secret key agreement.

The Key Distribution Protocols (KDPs) for UASNs were investigated in [26], [30], [31], [32], [33], [34], and [35]. The work in [30] demonstrated that in the context of highly constrained environments like UASNs, an efficient resolution to key distribution challenges is the application of a non-interactive identity-based key establishment protocol, like SOK [31]. Identity-based cryptography has been extensively used for designing KDPs for terrestrial Wireless Sensor Networks [32]. The main idea of this concept is to enable devices to compute other devices' public keys directly from

their identity information. This reduces the complexity and overhead associated with certificate management in Public Key Infrastructure (PKI) systems. However, a third-party entity called a Private Key Generator (PKG) is responsible for generating and securely distributing private keys to users within the system. This creates a single point of failure and raises concerns about key escrow as the PKG can decrypt all communications within the system. Moreover, identity-based cryptography often employs bilinear maps, which can be compromised with quasi-polynomial attacks on the discrete logarithm problem in small characteristic fields [33]. In [34], two data transmission protocols utilizing identity-based keying were proposed for mobile UASNs. One of them is based on bilinear mapping and the other is based on algebraic signatures. They are designed for underwater vehicles with different computation and storage costs and both enable direct authentication without external security infrastructure.

KDPs leveraging implicit certificates and Elliptic Curve Cryptography (ECC) were discussed in [26], [35], and [36]. Implicit certificates provide a means of certifying the relationship between a node's public key and its identity within the same data structure, thereby eliminating the need for an explicit signature. Due to their small size, they are a practical choice for key exchange in UASNs. In [35], the ECQV (Elliptic Curve Qu-Vanstone) protocol [37] was proposed for the distribution of implicit certificates in UASNs. For key exchange, two options were considered: FHMVQV (Fully Hashed Menezes-Qu-Vanstone) [38] and HOMQV [23] protocols. The combinations of these protocols were evaluated in terms of energy efficiency and time required to complete the exchange of keys between two nodes. The results showed numerous advantages of employing these protocols over the conventional Diffie-Hellman (DH) key exchange protocol, which is typically used with X.509 certificates. Notably, the combination of ECQV and HOMQV was found to be the most efficient, a conclusion that aligns with the findings from the study referenced in [36]. However, it is important to note that HOMQV is naturally vulnerable to replay attacks, an issue that these studies did not address.

Alternative approaches to authentication and key agreement in UASNs, which do not rely on (ECC) were proposed in [39] and [40]. Particularly, [39] suggested the use of chaotic maps to establish mutual authentication and negotiate session keys among users, a trusted gateway, and sensor nodes. This approach is distinguished by its computational efficiency. Meanwhile, [40] proposed a streamlined authentication scheme utilizing matrix addition instead of multiplication to minimize computational complexity. The proposed approach requires the assistance of a base station for generating configuration for each sensor node using Vandermonde vectors, which is the foundation for secret key generation. Additionally, the system employs a zero-knowledge proof protocol to ensure secure authentication without secret key enclosure, achieving significantly lower complexity compared to standard RSA-based methods. Both approaches

are based on a centralized trust model, which introduces a critical security vulnerability and could lead to performance bottlenecks in larger networks.

Besides cryptography-based authentication and key exchange solutions that are handled at higher layers of the protocol stack, many Physical Layer Security (PLS) methods were proposed. PLS authentication methods exploit the natural properties of underwater acoustic channels to authenticate the transmitter rather than using a cryptographic secret key. These properties are incredibly difficult to imitate, resulting in a strong authentication capability. Commonly used channel properties include: received signal strength (RSS) [41], power spectral density [42], wideband channel impulse response [43], [44], [45], [46], channel frequency response [47], and power level [48], [49]. More recent research explored various methods of generating keys by harnessing the inherent randomness of underwater acoustic channels. For example, multi-feature-based authentication was introduced in [47] where the distance, location, and arrival angle of the node were used. Channel impulse response (CIR) was used to generate keys in [50] and a database of CIRs in [51]. Machine learning approaches that fuse multiple channel features for facilitating authentication can be found in [52], [53], and [54] while the study in [55] utilized per-route packet time-of-arrival (ToA) and network topology as a shared source of randomness to generate keys. However, the application of PLS in dynamic, mobile environments is limited due to rapidly changing channel conditions. Additionally, the effectiveness of PLS in key generation depends on channel reciprocity, which cannot be guaranteed in all network topologies.

Although network clustering has proven effective in improving energy efficiency, scalability, and reducing communication delays in UASNs, the aspect of security in clustered networks remains largely unexplored. One of the pioneering efforts in this field is the key management protocol for mobile hierarchical networks described in [56]. This protocol centers around clustering sensor nodes around more capable nodes, integrating a Base Station (BS) for managing node identifiers and cryptographic elements, and acting as a gateway towards the terrestrial network. While the protocol demonstrated notable energy and storage efficiency, it does exhibit limitations, particularly in its reliance on preloading pairwise keys for sensor-cluster head communication and the absence of a key update mechanism. In [57], a similar model for secure authentication and data aggregation was proposed. The research presented in [58] followed a different approach where the BS issues re-authentication tickets to sensor nodes. These tickets can be validated by all CHs in the network, using a shared group key, which significantly streamlines cross-cluster authentication procedures in mobile scenarios. However, [58] mainly outlines the general concept of ticket-based authentication, lacking detailed discussion on the cryptographic algorithms involved and a plan for practical implementation.

B. MITIGATING INTERNAL ATTACKS

The reviewed papers effectively address message confidentiality, authentication, and integrity. However, they share a common vulnerability to internal attacks. Several types of internal attacks in UASNs, such as wormhole, sinkhole, and selective forwarding, were individually addressed in [59], [60], [61], and [62], respectively. Recently, reputation-based trust models were proposed as effective tools for simultaneously dealing with a wide range of internal attacks in UASNs [6], [63], [64], [65]. A trust mechanism can be applied when correct node behavior and misbehavior are defined precisely and can be detected. Nodes are trusted based on their past behavior within the network, which is reflected in their reputation scores. Although this strategy only identifies the attacker rather than limiting its effect, it can be applied to many different attacks, thus providing a general defense solution. In [63] and [64], the authors observed that the performance of malicious attacks is mainly affected by three factors: the state of communication, the abnormality of data packets, and the abnormality of energy consumption.

Recent literature, including a comprehensive survey conducted by [66], highlights the adoption of machine learning algorithms in wireless sensor networks to detect anomalous behavior. These approaches leverage advanced analytical techniques to enhance the detection and diagnosis of irregular activities within network operations.

C. BLOCKCHAIN-ASSISTED SECURITY

Blockchain technology is increasingly recognized for its potential to improve communication security in IoT environments [11], [67], including the specific domain of the IoUT [9], [10]. The focus of several studies was on resolving the single point of failure problem of conventional PKI that relies on a centralized CA for issuing, maintaining, and revoking public key certificates [13], [18], [21]. Blockchain features such as decentralization, traceability, and tamper-proof were exploited for the implementation of scalable and secure distributed PKI. In [18], blockchain was used to record key materials of devices in edge-computing assisted smart grids. A set of smart contracts was used to provide conditionally anonymous authentication of devices and support efficient key updates and revocation. In [68], authors proposed a blockchain-based security mechanism within the Clustered Edge Intelligence (CEI) framework that enables administrators and users to track and analyze immutable event chains, thus enhancing the capability to investigate any anomalies. A conditional privacy-preserving authentication protocol with implicit certificates for vehicular edge computing was proposed in [21]. In [13], smart contracts on a private blockchain platform were used to enable implicit certificate generation, access, and revocation. While the studies referenced in [13] and [21] proposed security solutions using blockchain and implicit certificates in edge computing infrastructures, which aligns with our approach, they did not address the unique challenges posed

by UASNs. Specifically, these studies ignored the significant propagation delays that make frequent communication with the blockchain time-consuming. Furthermore, these studies do not focus on simplifying the re-authentication process when devices move between edge servers.

A blockchain-based privacy-preserving authentication protocol and malicious node detection mechanism for IoUT were proposed in [9]. Similar to our approach, the authors utilize sink nodes on the surface as blockchain network for the registration and authentication of sensor nodes. During the registration phase, the blockchain stores hashes of the sensor node's credentials, which include the node's location, ID, and the owner's ID. For authentication, the sink node compares the stored hashes with newly generated ones during node login. If a match is found, the sensor node is granted network access. Additionally, the paper addressed the detection and partial revocation of suspected malicious nodes by integrating Weighted Trust Evaluation (WTE) [69] and Additive Increase Multiplicative Decrease (AIMD) [70] mechanisms. However, unlike our solution, this paper assumes a UASN model with static sensor nodes, as a change of node position would entail change of nodes' credentials. Furthermore, the paper did not address key exchange and management issues which are essential for maintaining network security.

The problem of cross-domain authentication has been explored in the context of industrial IoT [16], [17] vehicular networks [19], [20], [22], and drone networks [12], with blockchain used to establish trust among different domains. When it comes to UASNs, the application of blockchain for cross-cluster authentication has been investigated only in [8]. The approach proposed in [8] is certificates, with cluster heads generating the necessary key material for all nodes involved. Due to the absence of certificates, mutual authentication between sensor nodes necessitates interaction with a cluster head, which may significantly increase communication overhead and delay.

D. CONTRIBUTION

It is evident from the above discussion that many security approaches have been proposed for UASNs but limited works have been carried out concerning the decentralized authentication and key management in clustered UASNs. Keeping in view the necessity of providing security services with reduced latency and communication overhead while maintaining an adequate level of security, we designed a novel blockchain-based assisted authentication and key-management protocol with implicit certificates, which is not only secure but also lightweight and highly recommended in very constrained environments. While we rely on ECC cryptographic primitives to protect the network from a wide range of external attacks, we also exploit powerful security features of blockchain to mitigate vulnerabilities to reply attacks and internal attacks. Additionally, the decentralized nature of the proposed authentication protocol, leveraged by blockchain, allows for seamless movement of

authenticated sensor nodes between clusters without the need for re-authentication. This enhances network flexibility and operational efficiency, making it particularly suitable for dynamic UASN environments.

III. BEKMP: BLOCKCHAIN-BASED KEY MANAGEMENT PROTOCOL

In this section, we first outline the system and adversary models under consideration, then delve into the details of the proposed key management protocol.

A. NETWORK MODEL

We consider the problem of authentication in heterogeneous UASN, as shown in Fig. 1. Sensor Nodes (SNs) are randomly scattered underwater and collaboratively perform monitoring tasks. These nodes exhibit various mobility patterns—some drift passively with ocean currents, while others, such as Autonomous Underwater Vehicles (AUVs) or Remotely Operated Vehicles (ROVs), engage in intentional, directed movement. Despite the disparity in energy reserves and computational capacities among these diverse SN categories, enhancing energy efficiency remains a critical shared objective, as it directly impacts the operational duration of underwater sensing missions.

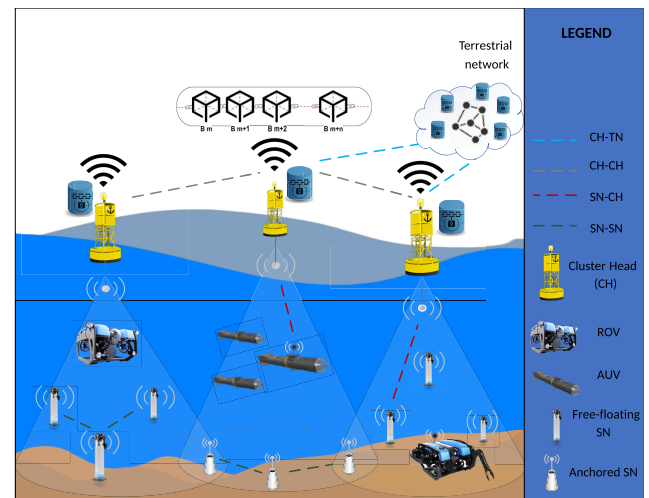


FIGURE 1. Considered UASN model.

SNs are organized in a number of clusters, each managed by a Cluster-Head (CH) deployed at the water surface. The CHs gather data from their respective cluster members, pre-process the information, and relay it to Terrestrial Network (TN) infrastructure for advanced analysis. Also, they handle device authentication and oversee node mobility. Network clustering yields many advantages, notably in terms of scalability, energy conservation, and reduced communication delays. By confining communication to within clusters, the transmission power requirements are substantially lowered — a critical consideration in the energy-constrained context of UASNs. CHs also act as data aggregation points, filtering and compressing information, thus reducing the data volume

that needs to be transmitted over the backhaul link to the TN. Moreover, the introduction of clusters streamlines network topology and routing, offering particular advantages in large-scale network settings.

The network's communication dynamics encompass several patterns: SN to CH, SN to SN, CH to CH, and CH to TN transmissions. The first two communication modes operate through underwater acoustic channels, characterized by their unique propagation challenges, whereas the latter two are facilitated by advanced radio communication technologies, which provide higher bandwidth and reliability in the air. Within this framework, we assume that CHs are not constrained in terms of energy, communication capabilities and computing power. To enable both fast and secure authentication of underwater SNs and establish trust within the clustered UASN, CHs are incorporated in a blockchain network that extends to the TN infrastructure. Authentication is addressed by means of asymmetric cryptography and ECQV certificates issued by a trusted CA. All CHs maintain an identical ledger that logs identity details, certificates, encrypted keys, and trust scores of SNs. To facilitate transparent and automated management of this data, three distinct smart contracts operate within the blockchain ecosystem:

- **Certificate Management Smart Contract:** This contract oversees the issuance and revocation of certificates, ensuring that SNs' certificate statuses remain up-to-date based on changes in trust scores and pre-defined expiration times.
- **Key Management Smart Contract:** Responsible for the secure storage, updates and retrieval of session-keys and managing key sharing rules.
- **Trust Management Smart Contract:** Designed to dynamically manage trust scores of SNs, allowing CHs to adapt to changing trust levels among SNs and automatically revoke certificates when necessary.

The blockchain network is private and permissioned. This means that each participant in the network must be authenticated and authorized. The authentication process prevents unauthorized nodes from joining the network and accessing the ledger. Such a system ensures a robust, decentralized approach to SN authentication and trust management within the network.

B. ADVERSARY MODEL

The UASN is exposed to various types of external attacks, necessitating robust defensive strategies. These attacks include, but are not limited to sybil, data tampering and reply attacks. Besides this, the network security mechanisms should provide protection against internal attacks originating from compromised SNs, such as sinkhole, wormhole, selective forwarding and exhaustion attacks [1]. We assume that only underwater devices can get compromised, primarily due to the fact that CHs have superior capabilities compared to SNs and their strategic placement at the water surface facilitates regular inspections. Consequently, CH to CH and CH to TN communication is presumed to be highly secure,

benefiting from the deployment of robust security mechanisms and advanced radio communication technologies that are less likely to be exploited by adversaries.

C. BEKMP PROTOCOL

The overall process of authentication and key management in the considered environment can be divided into the following five phases: 1) System Initialization Phase, 2) Registration Phase, 3) Mutual Authentication and Key Exchange Phase, 4) Service-Delivery Phase and 5) Trust Evaluation Phase. The detailed description of these phases is presented in the rest of the section, with some most frequently used notations summarized in Table 1.

TABLE 1. Notation summary.

Symbol	Definition
ID_x	Identifier of the entity x
N_x	Nonce value chosen by the entity x
P	Generator of the elliptic curve
n	Order of the elliptic curve
KDF	Key derivation function
H	Hash function
PK_x	Public key of x
sk_x	Private key of x
R_x	Ephemeral public key of x
r_x	Ephemeral private key of x
$Cert_{SN_x}$	Certificate of SN x
T_{exp}	Expiration time of the certificate
K_{x-y}	Session key shared by x and y
ENC_k	Encryption with key k
TR_e	Energy trust evidence
TR_d	Data trust evidence
TR_c	Communication trust evidence
E_{res}	Residual energy of SN
$ $	Concatenation operation

1) SYSTEM INITIALIZATION PHASE

In this phase, CA generates public parameters for the authentication procedures and then all SNs that participate in the environment monitoring mission should register with the corresponding CH for getting the blockchain-assisted authentication service. Both the ECQV protocol that is used for implicit certificate generation and the HOMQV protocol that is used for key-exchange rely on the ECC algorithm. Therefore, during system initialization, CA picks a large prime number n that defines an elliptic curve $E(n)$. Points in this curve form an additive cyclic group G , with P as the generator and n as the order. CA chooses its private key sk_{CA} randomly from the multiplicative group of integers modulo n , denoted as $\mathbb{Z}_n^*$. Then it derives a public key as $PK_{CA} = sk_{CA} \cdot P$, where $(\cdot)$ represents ECC multiplicative operation. CA also selects a one-way hash function (H) and Key Derivation Function (KDF) to be used during the Mutual Authentication and Key Exchange Phase. Finally, CA makes the parameters $\langle E, P, n, H, KDF, PK_{CA} \rangle$ publicly available.

During this phase, CHs are associated with the permissioned blockchain network and the identities of SNs are written to the blockchain ledger. As part of the blockchain

association process, each CH generates a private-public key pair. The public keys are certified by the CA, ensuring their authenticity and security. It is assumed that every CH in the network possesses knowledge of the public keys of all other CHs.

2) REGISTRATION PHASE

During this phase, underwater SNs register over a secure channel to get their private-public key pair. This phase is executed just once, and follows the procedure depicted in Fig. 2, which is explained in detail below:

Step 1: Initially, Sensor Node i (SN_i) selects a secret value $r_{SN_i} \in \mathbb{Z}_n^*$ at random. This secret value is used to compute a specific point on the elliptic curve, i.e. ephemeral public key represented as $R_{SN_i} = r_{SN_i} \cdot P$, where P is the generator point of the elliptic curve. Subsequently, SN_i transmits both the computed elliptic curve point R_{SN_i} and its unique identifier (ID_{SN_i}) to the designated CH node via a secure communication channel.

Step 2: Upon receiving the registration request from SN_i , the CH_j firstly invokes the *Certificate Management Smart Contract* on the blockchain to validate the node identity and check its registration status. If the received identity does not match any of the sensor node identities in the blockchain, the registration request is aborted. The same happens if CH finds a record in the blockchain that indicates that the node is already registered. This step is important for preventing sybil and replay attacks. If the request passes the validation procedure, the smart contract issues a certificate generation request via calling the off-chain CA service.

Step 3: On request, the CA service constructs an implicit certificate using the ECQV protocol. The ECQV protocol utilizes complex mathematical properties to link identity information and public keys into certificates. Unlike traditional methods, it does not include the sender's public key in the certificate but instead allows the recipient to calculate it using the sender's certificate and the CA's public key, thereby reducing communication overhead. The process of certificate generation includes several steps. Firstly, the CA randomly selects a secret value $r_{CA} \in \mathbb{Z}_n^*$ and computes an ephemeral public key $R_{CA} = r_{CA} \cdot P$. Then it produces the certificate $Cert_{SN_i} = (R_{CA} + R_{SN_i}) || ID_{SN_i} || T_{exp}$, which consist of a point on the elliptic curve followed by the node ID and the certificate expiration date T_{exp} . The CA also generates a key reconstruction material [37]:

$$c = sk_{CA} + H(Cert_{SN_i}) \cdot r_{CA} \mod n \quad (1)$$

Finally, CA forwards the certificate and c to the CH that sent the request for certificate generation.

Step 4: CH forwards the response from the CA service to SN_i . Then, it embeds the implicit certificate into a transaction and adds it to the blockchain. The purpose of this step is to eliminate the need for sending the certificate during SN-CH communication.

Step 5: Upon getting the response from the CH, SN_i derives its long-term private and public keys in accordance with the

ECQV protocol [37], as follows:

$$sk_{SN_i} = c + r_{SN_i} \cdot H(Cert_{SN_i}) \mod n \quad (2)$$

$$PK_{SN_i} = sk_{SN_i} \cdot P \quad (3)$$

The received certificate is validated by checking if the following equality holds:

$$\begin{aligned} PK_{SN_i} &= sk_{SN_i} \cdot P \\ &= PK_{CA} + Cert_{SN_i} \cdot H(Cert_{SN_i}) \end{aligned} \quad (4)$$

If all the required steps are completed successfully, SN_i obtains a valid certificate enabling identity verification within the network. Whenever a CH triggers an update in a smart contract to modify the certificate of one of its nodes, the other CHs will be notified via the blockchain network, so they can update their respective cached ledgers accordingly.

3) MUTUAL AUTHENTICATION AND KEY EXCHANGE PHASE

During this phase, each SN initiates mutual authentication with its intended communication peer, either a CH or another SN. This is achieved by employing the HOMQV protocol, known for its efficiency and one-way operational design. The protocol facilitates the exchange of a session key using just a single message, streamlining the authentication process. This feature is particularly advantageous in UASNs as it conserves energy for the SNs. It is important to highlight that this authentication procedure differs depending on whether the communication involves an SN and a CH or occurs between two SNs.

a: SN-CH COMMUNICATION

The overall process of SN-CH authentication and key-exchange is visually represented in Fig. 3 and will be thoroughly explained below:

Step 1: To initiate the process, SN_i transmits an authentication request containing its ID and a randomly generated nonce value N_{SN_i} to its corresponding cluster head CH_j .

Step 2: Upon receiving the authentication request, CH_j firstly invokes the *Key Management Smart Contract* on the blockchain to determine if SN_i is already authenticated and has a valid session key. If SN_i is already authenticated with valid key material, the request is rejected. The request is also rejected if the nonce value is determined to be invalid, such as in cases where a request with an identical nonce value has already been processed. If the key exchange has not yet occurred, lacks confirmation from SN_i , or the key has expired, CH_j retrieves SN_i 's implicit certificate from its ledger and extracts SN_i 's public key using the calculation outlined in 4. Then, CH_j generates an ephemeral key pair as $R_{CH_j} = r_{CH_j} \cdot P$, where $r_{CH_j} \in \mathbb{Z}_n^*$ is a random positive integer within the range $[1 \dots n - 1]$. Following this, CH_j computes the shared session-key using the KDF in line with the HOMQV protocol [23], as follows:

$$K_{SN_i-CH_j} = KDF(\sigma, ID_{SN_i}, ID_{CH_j}, R_{CH_j}, N_{SN_i}) \quad (5)$$

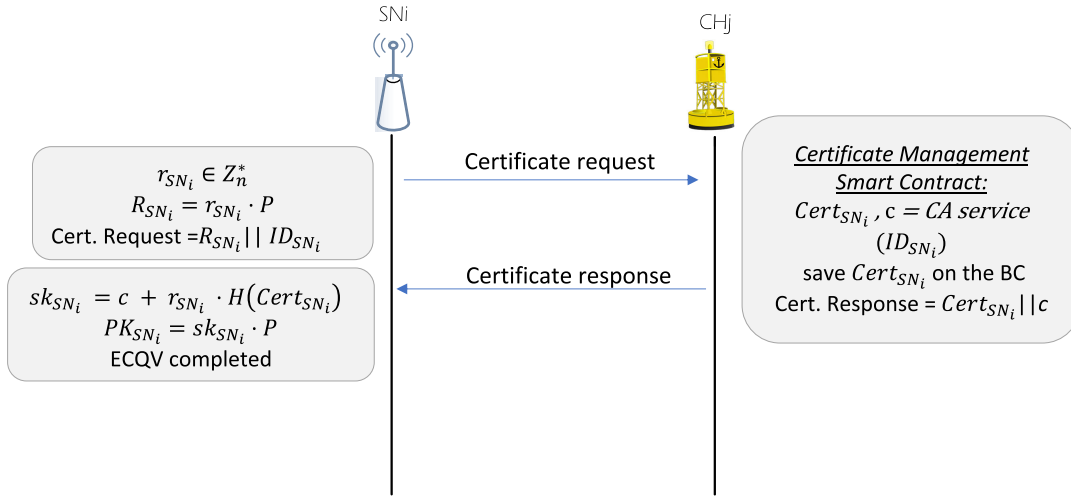


FIGURE 2. Registration phase.

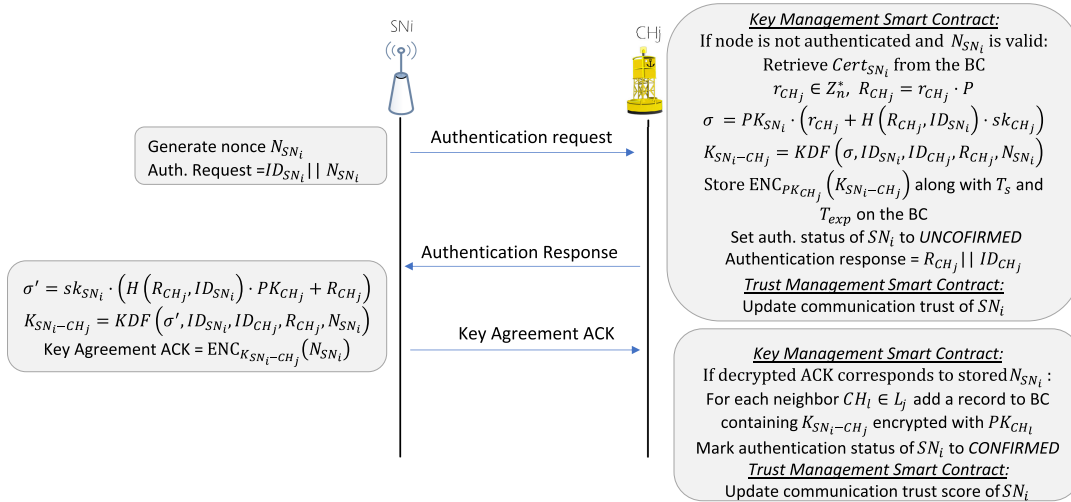


FIGURE 3. SN-CH authentication and key-exchange.

where \$\sigma\$ is:

$$\sigma = PK_{SN_i} \cdot (r_{CH_j} + H(R_{CH_j}, ID_{SN_i}) \cdot sk_{CH_j}) \mod n \quad (6)$$

Subsequently, \$CH_j\$ creates a blockchain record in the form of a key-value pair: \$(ID_{CH_j}, ID_{SN_i}, N_{SN_i}) \rightarrow (T_s, ENC_{PK_{CH_j}}(K_{SN_i-CH_j}), T_{exp}, confirmed = False)\$. This record indicates that a session-key for \$SN_i - CH_j\$ communication has been generated based on the nonce \$N_{SN_i}\$, but has not been confirmed as valid by \$SN_i\$, as indicated by the last element of the entry. Here, \$T_s\$ represents timestamp of the record, \$ENC_{PK_{CH_j}}\$ denotes the encryption operation with \$CH_j\$'s public key, and \$T_{exp}\$ represents the session key's expiration time. Finally, \$CH_j\$ replies to \$SN_i\$ with the values of \$R_{CH_j}\$ and \$ID_{CH_j}\$.

It is important to note that multiple authentication records with various nonce values may concurrently exist in the

blockchain ledger, as a result of malicious attacks. However, only \$SN_i\$ can derive the session key from \$CH_j\$'s response. Once \$SN_i\$ confirms key derivation, the specific record that corresponds to the nonce value selected by \$SN_i\$ is designated as valid, while the rest are marked as deleted or invalid. Additionally, the timestamps of all records associated with unconfirmed key-exchange attempts will undergo continuous review, with any outdated entries being labeled as invalid.

Step 3: \$SN_i\$ generates the session key using Equation 5, computing \$\sigma'\$ as follows:

$$\sigma' = sk_{SN_i} \cdot (H(R_{CH_j}, ID_{SN_i}) \cdot PK_{CH_j} + R_{CH_j}) \mod n \quad (7)$$

This step ensures that both \$SN_i\$ and \$CH_j\$ possess an identical session key, as \$\sigma\$ equals \$\sigma'\$. To acknowledge the receipt of the packet, \$SN_i\$ sends an acknowledgment to \$CH_j\$ containing \$N_{SN_i}\$ encrypted with the session key.

Step 4: In response to the acknowledgment received from SN_i , CH_j adds new entries in the blockchain. Each entry corresponds to a neighboring cluster head $CH_l \in L_j$, where L_j represents the set of all neighboring CHs for CH_j . The format of these entries is as follows:

$$\begin{aligned} &\langle ID_{CH_l}, ID_{SN_i}, N_{SN_i} \rangle \\ &\rightarrow (T_s, \text{ENC}_{\text{PK}_{CH_l}}(K_{SN_i-CH_l}), T_{exp}, \text{confirmed} = \text{True}) \end{aligned} \quad (8)$$

These records serve two purposes: they prevent replay attacks in the HOMQV protocol and eliminate the need for re-authenticating SNs during cross-cluster migrations.

b: SN-SN COMMUNICATION

The establishment of a secure communication channel between SNs is done without interaction with the blockchain. In this case, SNs are required to directly exchange their certificates. While the blockchain adds an extra layer of security to the network, we have decided on this approach because of the significant propagation delays inherent in UASNs. Introducing blockchain interaction for authentication status checks and certificate retrieval would introduce considerable communication delays. Therefore, we have opted for an approach in which SNs directly exchange ECQV certificates and rely solely on HOMQV for authenticated key exchange.

4) SERVICE DELIVERY PHASE

During the Service Delivery Phase, SNs periodically report their measurements and battery status to the corresponding CHs at regular intervals. If SN_i , either intentionally or due to external factors such as sea currents, migrates to a new cluster, it is required to submit an association request to the new cluster head CH_l , $l \in L_j$. In response, CH_l activates the *Key Management Smart Contract* to verify the authentication status of SN_i . This involves searching the ledger for a key-value pair where the key contains both ID_{SN_i} and ID_{CH_l} . If such a pair is found, CH_l deduces that SN_i is trustworthy and already authenticated, thereby eliminating the need for a new session-key generation process. Instead, CH_l extracts the encrypted session-key, $\text{ENC}_{\text{PK}_{CH_l}}(K_{SN_i-CH_l})$, from the blockchain ledger. This session-key is then decrypted using CH_l 's private key, sk_{CH_l} . An exception arises if the session key is found to be expired, in which case the key-value pair is marked as invalid. Furthermore, CH_l is responsible for updating the blockchain with records that claim SN_i 's authenticated status across neighboring clusters $CH_{l'}$, $l' \in L_l$. These records, formatted as $\langle ID_{CH_{l'}}, ID_{SN_i}, N_{SN_i} \rangle \rightarrow (T_s, \text{ENC}_{\text{PK}_{CH_{l'}}}(K_{SN_i-CH_{l'}}), T_{exp}, \text{True})$ include the encrypted session-key. In this way, the system eliminates the need for SN_i to undergo re-authentication each time they traverse different clusters, significantly streamlining the process and ensuring continuous, secure operation within the network. The summary of this phase is illustrated in Fig. 4, providing a visual representation of the key elements and processes involved.

5) TRUST EVALUATION PHASE

To protect against internal attacks from compromised SNs, the proposed system provides Trust Management Smart Contract within the blockchain service layer. This contract assigns a trust score to each node, which is updated by CHs each time they receive a packet from SNs. The smart contract is pre-configured with a specific threshold for the trust score. If a node's trust score falls below this threshold, the system automatically initiates a certificate revocation process. This mechanism ensures accurate and transparent management of trust evaluations and updates.

Although UASNs face diverse attacks, the consequences of these attacks are primarily reflected in three aspects: communication failure, increased energy consumption, and data packet tampering [63], [64]. Therefore, as proposed in [63] and [64], our system uses three types of trust evidences to compute trust scores of SNs:

- **Energy evidence TR_e** - serves as a crucial indicator in detecting whether an SN is compromised and involved in energy-intensive attacks like exhaustion attack, or in activities that lead to lower energy usage, such as selective forwarding attacks. We assume that SNs report the residual energy level in each data packet sent. Energy trust is assessed by comparing a node's residual energy (E_{res}) against a predefined threshold, and its energy consumption rate (E_{cr}) against a historical average. It decreases as the deviation of the node's energy consumption from E_{cr} increases. Furthermore, if E_{res} falls below the set threshold, the energy trust is deemed zero.
- **Data trust TR_d** - based on the premise that packets transmitted by neighboring nodes in UASNs usually have similar values and follow a normal distribution [63]. The CH calculates the data trustworthiness of a node by comparing the sensor measurements from that node to the average of measurements reported by its neighbors. A larger deviation results in a lower trust value.
- **Communication trust TR_c** is assessed using a Subjective Logic framework, representing trust as a triplet of belief, disbelief, and uncertainty [63]. It is calculated by CHs based on the number of successful and unsuccessful communications with a particular SN. This model takes into account that packet loss might occur both due to the unreliability of the acoustic channels and the potential malicious behavior of nodes.

The trust evaluation method, while crucial, is beyond the scope of this paper. For a detailed understanding of the trust evidences used, interested readers are directed to [63] and [64]. In this paper, we employ a straightforward approach to trust scoring, calculating it as a weighted sum of the previously described trust evidences. If SN's trust score falls below the pre-defined threshold value, the certificate stored in the blockchain ledger will be marked as revoked. Following this, CHs disseminate details of the revocation to all SNs. The permanence of the blockchain database ensures that information regarding all revocation events is securely

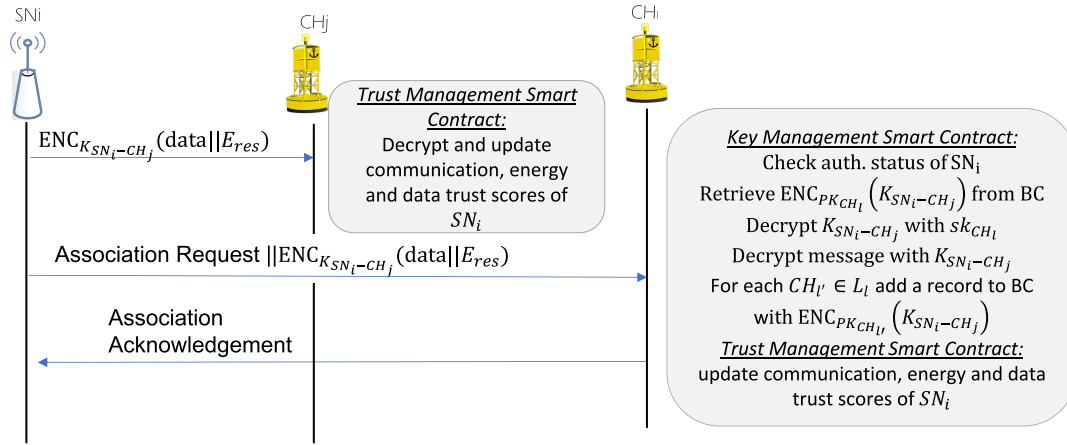


FIGURE 4. Service delivery and cross-cluster switching.

stored, providing a valuable resource for future reference and analysis.

IV. SECURITY EVALUATION

In this section, we provide a comprehensive analysis of BEKMP's security features, followed by a detailed formal verification using the AVISPA tool [24].

A. SECURITY ANALYSIS

In the proposed solution, the authenticated key exchange process is based on the ECQV and HOMQV protocols. This approach fundamentally relies on the complexity of two cryptographic problems: the Elliptic-Curve Discrete Logarithm Problem (ECDLP) and the Computational Diffie-Hellman Problem (CDHP) [71]. In our practical implementation, we utilized the SECG-P192 Elliptic Curve, offering a 96-bit security level. This implies that a malicious entity would need to execute approximately 2^{96} operations to compromise the discrete logarithm problem [72].

HOMQV provides robust mutual authentication, where both parties in the communication can authenticate each other's identities. This feature ensures that the entities involved in the key exchange are who they claim to be, reducing the risk of impersonation attacks. A unique session key is generated for each communication session. This key is derived from both public and private key components, which enhances its security against various cryptographic attacks. The protocol is designed to ensure that the disclosure of one session key does not compromise other session keys or the private keys of the participants. The compromise of ephemeral private keys does not compromise the session keys of past communications. This is because the session keys are derived using a combination of ephemeral and long-term keys.

While inheriting many security attributes of the Diffie-Hellman protocol, HOMQV significantly enhances computational efficiency by reducing the number of cryptographic operations. However, this efficiency compromises Perfect

Forward Secrecy (PFS). PFS ensures that the confidentiality of past communications remains secure even if long-term private keys are later compromised. In contrast, HOMQV's design involves partial derivation of session keys from the sender's long-term private key, violating the principle of PFS where session keys should be entirely ephemeral. This means that if the sender's key is compromised, previously secured sessions could potentially be decrypted. This limitation is a result of HOMQV's approach to not require the receiver to generate a new ephemeral key pair for each session, which, while reducing computational load, increases vulnerability. Nevertheless, such design represents a strategic balance between efficiency and security, especially beneficial in resource-constrained environments such as UASNs, where short-term secrecy suffices for operations like target detection and tracking, underwater surveillance, naval tactical communication, etc. For environments where long-term security is paramount, Full HMQV (FHMV) [38] offers a more secure alternative. Although more resource-intensive, FHMV adheres to the principles of PFS, making it a preferable choice for scenarios demanding robust long-term confidentiality.

Another inherent vulnerability of the basic HOMQV protocol is susceptibility to replay attacks [73]. To mitigate this, our implementation incorporates a nonce mechanism. Initially, the sender transmits a nonce unencrypted as part of the authentication request. Subsequently, after the session key is derived, this nonce is encrypted, as detailed in Section III. We also address the challenges posed by the highly unreliable underwater acoustic channel, which can lead to scenarios where an authentication request replied by an attacker is received before the original request, due to packet loss. Furthermore, during the second step of authentication, an attacker may initiate a replay attack, compelling the receiver to generate a new key for a legitimate node. While the attacker cannot derive this key, such attacks can disrupt communication with valid nodes. To safeguard SN-CH communications against these issues, we log authentication

requests in a blockchain ledger. Authentication is deemed complete only after receiving the encrypted nonce back from the valid sender of the authentication request.

The ECQV component of the proposed authentication method enables efficient verification of node identities by generating implicit certificates. While in conventional systems with PKI based on X.509 certificates, nodes need to prove knowledge of their private keys to the CA to avoid unauthorized public key certification, ECQV certificates remove the need for such proof, as there is no pre-set public key before issuance. The CA's involvement in the process ensures that nodes cannot influence their public key, thereby preventing malicious influence of any node. Different from X.509 certificates, implicit certificates do not incorporate a digital signature. Although one can theoretically select any identity and a random value to form a certificate, without the CA's participation, calculating the corresponding private key for the public key generated from the certificate becomes practically infeasible.

Besides authentication, the proposed security framework provides confidentiality and traceability as key features. Throughout the entire process, sensitive private information is consistently encrypted using shared keys. The behavior and authentication activities of sensor nodes are continuously tracked and recorded in the permissioned blockchain ledger, which is only accessible to authorized users. If any misbehaviors are detected, the certificate and corresponding public key of the node are automatically revoked by the *Trust Management Smart Contract*. All the records in the blockchain ledger are validated through consensus mechanisms, which involve all blockchain nodes. The records are encrypted, and the blockchain guarantees their immutability and tamper-proof nature, which ensures that traceability can be guaranteed. Additionally, the decentralized blockchain infrastructure makes authentication services provided by CH resistant to Denial of Service (DoS) attacks.

Finally, we compare our scheme with some recent research related to IoT and IoUT authentication, in terms of the supported security features, as Table 2 shows.

B. FORMAL VERIFICATION BASED ON AVISPA

To perform a formal security verification of the proposed mutual authentication protocol, we used the well-known AVISPA tool [24]. This open-source suite aids in the analysis and verification of security protocols against a broad spectrum of established attacks, operating under the Dolev-Yao (DY) threat model [75]. Before a protocol can be processed by AVISPA, it must first be defined using the High-Level Protocol Specification Language (HLPSL). HLPSL structures its syntax around 'roles', which represent the participants in the protocol, and 'composition roles', which are used to define sessions and the environment in which the protocol operates. This approach allows for a clear and structured representation of the protocol's functionalities and interactions between its various components.

In our AVISPA simulation, we allow the intruder's knowledge to include each node's ID, public keys, generator point, hash functions as well as all exchanged messages. To ensure security, two key objectives are defined. The first, pertaining to secrecy, necessitates maintaining the confidentiality of the shared key among the parties involved in the communication. The second, focusing on authentication, demands that the receiving parties validate the integrity of both nonces and ephemeral keys. AVISPA integrates four back-ends that apply diverse state-of-the-art automatic analysis techniques: "On-the-fly mode-checker (OFMC)", "Constraint-logic-based Attack Searcher (CL-AtSe)", "SAT-based Model Checker (SATMC)", and "Tree Automata based on Automatic Approximations for the Analysis of Security Protocols (TA4SP)" [24]. The back-ends help validate the safeness of the proposed mechanism against the targeted security objectives and provide the user with a detailed trace in case of violation.

We tested the proposed mutual authentication method on the OFMC and CL-AtSe backends. The simulation results confirmed that the proposed authentication mechanism is safe in both cases, as can be seen from Fig. 5.

File	File
SUMMARY	% OFMC
SAFE	% Version of 2006/02/13
DETAILS	SUMMARY
BOUNDED_NUMBER_OF_SESSIONS	SAFE
TYPED_MODEL	DETAILS
PROTOCOL	BOUNDED_NUMBER_OF_SESSIONS
/home/span/span/testsuite/results/homqv.if	PROTOCOL
GOAL	/home/span/span/testsuite/results/homqv.if
As Specified	GOAL
BACKEND	as_specified
CL-AtSe	BACKEND
STATISTICS	OFMC
Analysed : 64 states	COMMENTS
Reachable : 64 states	STATISTICS
Translation: 0.00 seconds	parseTime: 0.00s
Computation: 0.03 seconds	searchTime: 0.14s
	visitedNodes: 266 nodes
	depth: 6 plies

FIGURE 5. AVISPA simulation results.

V. PERFORMANCE EVALUATION AND COMPARISON

In this section, we assess the efficiency of the proposed BEKMP authentication and key-management protocol by examining computation and communication overheads, and energy efficiency. Firstly, we present the results of blockchain performance analysis, focusing primarily on the response times of smart contracts across various operations. Subsequently, we detail the implementation of the protocol in actual UASN and the DESERT Underwater simulator [76], discussing outcomes from experimental trials alongside simulation results for larger-scale UASN deployments. During these evaluations, the effectiveness of BEKMP is benchmarked against a closely related scheme referenced in [22], providing a comparative analysis of its operational efficiency.

For our key management system, we leveraged the Hyperledger Fabric SDK [77] to build a permissioned blockchain platform. To efficiently handle elliptic curve operations, we used the lightweight microECC library [78], specifically

TABLE 2. Comparison of security features across various schemes.

Features	Schemes									
	[35]	[74]	[25]	[13]	[21]	[22]	[8]	[18]	[9]	BEKMP
Cross-domain re-authentication	-	-	-	-	-	✓	✓	-	-	✓
Confidentiality	✓	✓	✓	-	✓	✓	✓	-	✓	✓
Mutual authentication	✓	✓	-	-	✓	✓	-	-	✓	✓
Lightweight design	✓	✓	✓	✓	-	-	✓	-	✓	✓
Access revocation	-	-	-	✓	-	-	-	-	✓	✓
Insider attack protection	-	-	-	-	-	-	-	-	✓	✓
Replay attack protection	✓	✓	✓	-	✓	✓	✓	-	✓	✓
Denial of service protection	-	-	-	-	✓	✓	✓	-	✓	✓

designed for resource-constrained devices. We chose the SECG-P192 Elliptic Curve, striking a balance between efficiency and security. This implies a robust 96-bit security level, 24-byte long private keys and 48-byte long public keys. For enhanced security, we integrated the SHA-256 hashing function alongside the HMAC-SHA256 key derivation function. To streamline communication, we allocated 1 byte for message type identification, 3 bytes to uniquely identify each node, and 4 bytes for nonce values. Our implementation also adheres to the following design principles and best practices:

- **Cryptographic Operations:** Critical cryptographic operations are executed without conditional branches, employing time-constant code to mitigate the risk of timing side-channels [79].
- **Memory Management:** Variable allocation is handled through static memory allocation methods, avoiding the use of dynamic memory allocation techniques.
- **Data Transmission:** Elliptic curve points are transmitted in a compressed format to reduce network overhead, adhering to standards outlined in SEC 1 [80]. This approach results in a compact certificate length of just 32 bytes, where 25 bytes are dedicated to the elliptic curve point, 3 bytes for identification, and 4 bytes for T_{exp} . This strategic optimization ensures that our payload transmissions remain comfortably within the maximum payload limit of 48 bytes (without Base64 encoding overhead) supported by the NMV3 acoustic modems [81].

A. BLOCKCHAIN PERFORMANCE ANALYSIS

The simulations were conducted on an x86_64 GNU/Linux system equipped with a single virtual CPU (1.7 GHz) and 512 MB of RAM. Using Docker Engine, we deployed a blockchain network comprising $N_b \in \{3, 5, 10\}$ permissioned nodes. In line with Hyperledger Fabric's architecture, our blockchain network consisted of two types of blockchain nodes: Endorsers and Orderers. Each CH in our configuration acted as an Endorser, responsible for hosting a copy of the blockchain ledger, executing smart contracts, and validating proposals of data transactions. In addition to these, we assigned the role of Orderer to one CH, tasked with compiling all endorsed transactions into new, sequentially ordered blocks, thereby ensuring the network's integrity and consistent transaction processing.

To assess the efficiency of smart contracts, we simulated underwater SNs with multi-threaded software. In these simulations, the number of threads, representing the simulated nodes, varied according to a Poisson distribution process. The mean rate parameter of this process, denoted as λ , was systematically varied from 1 to 30 transactions per second (tps), enabling an in-depth analysis of smart contract performance under varying loads. The evaluation process involved a step-by-step examination of the key smart-contract functions, including: certificate generation, key generation, key access, and trust score update. The results of these evaluations are presented in Fig. 6 and discussed below.

1) CERTIFICATE GENERATION

We measure the time taken from the moment when node identity and other necessary parameters are submitted as part of the request until the implicit certificate is produced. This process includes validating the node ID on the blockchain and executing the CA service. The results in terms of response time are presented in Fig. 6(a). As expected, interactions with the blockchain lead to an increase in response time. However, this additional cost is justifiable, considering that this operation is infrequently triggered, and the security benefits introduced by blockchain outweigh the added delay.

2) CERTIFICATE ACCESS

Accessing certificates is a crucial function as it is used during every key exchange. As shown in Fig. 6(b), the average retrieval time is relatively small. The utilization of implicit certificates not only minimizes storage requirements but also significantly enhances access speed.

3) KEY CREATION AND KEY ACCESS

The key generation process employs the HOMQV protocol. Fig. 6(c) illustrates the time required for key generation, comparing scenarios with and without blockchain integration. In the blockchain-integrated scenario, the results presented encompass several steps, starting from the reception of the key generation request. These steps include accessing the blockchain for certificate verification, key derivation, and the storage of the encrypted key on the blockchain to enable cross-cluster re-authentication. The encryption step is a significant factor contributing to the observed increase in

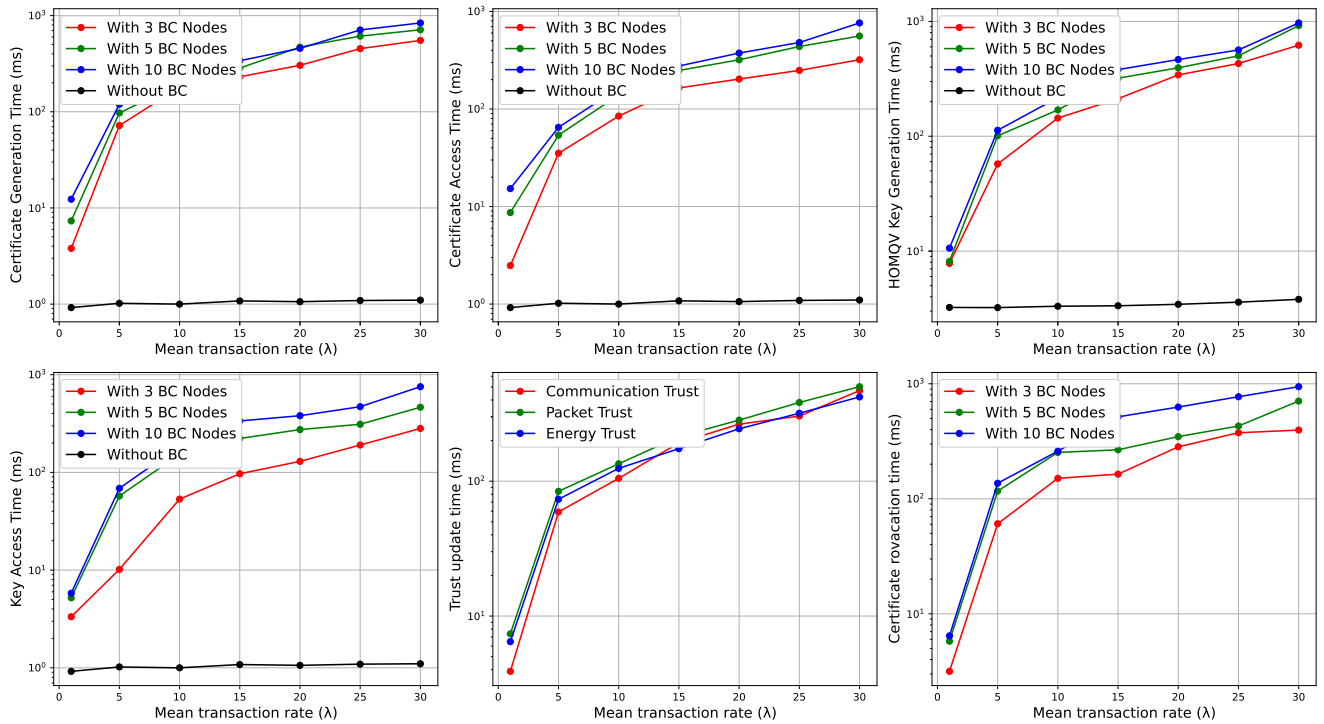


FIGURE 6. Simulation analysis of blockchain (BC) network performance with varying numbers of nodes and under variable load conditions.

processing time. The results regarding the retrieval time of the encrypted key are depicted in Figure 6(d).

4) TRUST MANAGEMENT

Trust evidences are continuously updated and recalculated upon receiving packets or detecting packet loss. Information about the last 50 energy, communication, and data evidences are securely preserved in an encrypted format on the blockchain. In cases where a threat is identified based on the total trust score of a node, the node's certificate is automatically revoked. The time required to trigger trust update operations through smart contract functions is depicted in Fig. 6(e). Generally, all trust update operations demonstrate comparable efficiency, with data trust processing being the most time-consuming. The timing details for the certificate revocation operation are illustrated in Fig. 6(f). The revocation process involves retrieving all three trust evidences from the blockchain and then performing a weighted sum calculation to determine the overall trust score, which is used as a basis for certificate revocation.

B. PERFORMANCE ANALYSIS OF BEKMP PROTOCOL IN UASN

We conducted a comprehensive analysis of the BEKMP protocol's performance in UASN environments, utilizing both simulations and experimental methods. Initially, we performed experiments on real devices to gather essential data, which informed and enhanced the realism of the subsequent simulations.

1) EXPERIMENTAL RESULTS

To further validate the proposed security framework in a practical context, particularly its time efficiency in an underwater setting, we conducted a series of sea trials using a small-scale UASN. The trials took place during the Breaking the Surface workshop [82] in Kumbor, Montenegro. The network setup for these trials included an Arduino Mega 2560 micro-controller board, which functioned as an underwater SN, along with three aPad surface vehicles [83], serving as CHs in the network (Fig. 7). Each device in this configuration was equipped with the NMV3 underwater acoustic modem [81] to enable underwater communication. In addition, the aPads were equipped with long-range WiFi antennas for over-the-air communication.

We evaluated the end-to-end latency of executing authentication using our security framework. For a comparative perspective, we implemented the reference algorithm detailed in [22] on identical hardware and tested under the same scenario. We measured the time required for each critical phase of the authentication protocols on the underwater SN and presented the results in Tables 3 and 4. Additionally, we evaluated the computational load on the CHs, and these results are shown in Tables 5 and 6.

It is important to note that the underwater acoustic modems employed in our experiments support a maximum packet payload size of 64 bytes. Due to this constraint, the certificate issuance process in proposed protocol was divided into two parts: (i) sending the ECQV certificate, and (ii) transmitting the key reconstruction material c . Similarly, the registration



FIGURE 7. Experiment in the adriatic sea with aPads as cluster heads.

TABLE 3. Time costs of BEKMP operations.

BEKMP Operations	Time Cost (ms)
Certificate request generation	1136
Total time for certificate acquisition	6746
Extraction of public key	1077
Certificate validation time	2155
Key reconstruction material retrieval (c) and long-term key computation	6460
Authentication time	10924

TABLE 4. Time costs of authentication protocol proposed in [22].

[22] Operations	Time Cost (ms)
Secret key acquisition time	11302
Public key acquisition time	6750
M retrieval time	6440
Authentication time	23649

TABLE 5. Time costs of BEKMP blockchain (BC) operations.

BEKMP BC Operation	Time Cost (ms)
Verification of SN identity and status	7.2
Certificate creation time	1.04
Certificate storing time	1.95
Key generation time	3.44
Key storing time	5.92
Key retrieval time	1.55

TABLE 6. Time costs of BC operations in [22] protocol.

BC Operation in [22]	Time Cost (ms)
Validation of device	7.32
Device registration	3.09
L storing time	1.94

process for the protocol proposed in [22] also necessitated multiple packet transmissions to fit within the modem's payload limit. In Tables 3 and 4, *Authentication time* is defined as the cumulative duration required for a node to complete its authentication process, starting from the moment the initial authentication request is sent to the cluster head. Clearly, the proposed method achieves faster authentication by utilizing a one-way key exchange, in contrast to the two-way exchange used [22]. As can be observed from Tables 5 and 6, the computational demands on CHs were found to have a minimal impact on the overall delay. This

is because aPad devices have relatively high processing capabilities.

During experiments we often faced challenges with regard to environmental interference, which occasionally affected the transmission efficiency of communication between the SN and CHs. These factors sometimes led to data loss and additional delays. However, to ensure fair comparison, the time cost of the authentication protocols is evaluated only based on measurements related to sessions that did not involve any packet loss in the authentication phase.

When evaluating computational complexity, we can compare the BEKMP protocol with the reference protocol [22] by focusing on the number of times they execute cryptographic hash functions (N_h) and perform ECC-based scalar multiplications (N_{ecc}) during the registration and authentication phases. The BEKMP protocol leverages the ECQV protocol during registration, which entails generating an ephemeral key pair, a certificate, and validating this certificate. This process involves three scalar multiplications and three hash computations. We consider the time spent on XOR operations and concatenations as negligible due to their minimal computational impact. The registration process via [22] involves a simpler mechanism where a central trusted authority generates a secret/public key pair for all SNs. Despite its higher computational demand, particularly in certificate generation and validation, the ECQV protocol used in BEKMP offers significant advantages for SN-SN communication. Once SNs are registered and equipped with their ECQV certificates, they can securely communicate with each other by directly exchanging certificates. This significantly streamlines network operations and reduces communication overhead. For the authentication phase, the computational overhead of the reference protocol [22] is quantified as $5 \cdot N_h + 6 \cdot N_{ecc}$. In contrast, BEKMP offers a reduced overhead of $2 \cdot N_h + 4 \cdot N_{ecc}$.

In Table 7, we provide a detailed comparison of the communication overheads. Our security scheme requires a total of 140 bytes for registration, while the reference protocol [22] demands 164 bytes for the same process. Utilizing the HOMQV one-pass method, our scheme cuts down the communication overhead for SN-CH authentication to just 64 bytes. However, in mutual authentication scenarios between two SNs, this requirement increases to 104 bytes, primarily due to the transmission of certificates. In contrast, the protocol from [22] involves a more substantial exchange of messages, resulting in an overall communication overhead of 204 bytes for completing authentication, regardless of whether it is between SNs or between an SN and a CH. It is important to note that the overall communication overhead is significantly influenced by the packet size limitation of underwater acoustic modems, as this constraint directly influences the total number of packets exchanged.

A notable advantage of our proposed security scheme is its efficiency in cross-cluster authentication, where it avoids any additional communication and computational overhead. This efficiency stems from the fact that CHs obtain the shared

TABLE 7. Comparison of communication overheads.

Operation	BEKMP (in bytes)	[22] (in bytes)
Registration request	48	36
Registration response	92	128
SN-CH Authentication	64	204
SN-SN Authentication	104	204

key directly from the blockchain network, unlike the [22] approach, which requires two additional hash operations followed by transmission of two additional messages, adding up to 40 bytes, during cluster handover.

The analysis conclusively shows that our proposed security scheme significantly reduces the authentication latency, which is a result of reduced communication and computation overhead.

2) UASN SIMULATION RESULTS

To evaluate the effectiveness of the proposed BEKMP protocol and its comparative efficiency against the protocol referenced in [22] in larger UASNs, we utilized the DESERT Underwater simulator [76]. We considered three UASN configurations with 6, 12, and 25 SNs deployed within a 2 km x 2 km x 600 m volumetric underwater area. Three CHs are strategically positioned at fixed coordinates: (1000, 333, 0), (666, 1666, 0), and (1333, 1666, 0), ensuring comprehensive coverage. The total duration of the simulation was set at 28 hours.

In the simulation setup, both SNs and CHs were configured with acoustic modems operating at a carrier frequency of 24 kHz and a bandwidth of 5 kHz. These modems support a bitrate of 640 bps and a transmit power of 160 dB re uPa, with a reception power consumption of 0.62 W. This configuration mirrors the specifications of NMV3 acoustic modems utilized in our actual field deployment, providing a realistic basis for evaluating the protocol's performance at scale.

SN mobility is simulated using the built-in *uwdriftposition* module of the DESERT Underwater simulator [84]. This module assigns each SN a deterministic speed vector, which is further modified by time-varying “noise” vectors to accurately represent the influence of ocean currents on SN movement. The module continuously updates the node's location based on the current's direction and speed. Each update incorporates random noise to mimic the undulating motion typical of objects adrift in water. By introducing these dynamic elements to the network topology, the module effectively simulates the impact of sea currents and supports realistic scenarios of cross-cluster communications.

We evaluated the protocols under the assumption that all SNs establish a secure channel only with the CHs. The data transmitted by each SN are generated according to a Poisson distribution with a mean rate of 1 packet per minute. SN sends a packet to a CH as soon as the channel is considered free, utilizing the Carrier Sense Multiple Access (CSMA) MAC protocol to minimize the risk of collisions. We assume that communication between SNs and CHs is always conducted over direct links, i.e. without the intervention of relay nodes

and routing protocols. Packet payloads are varied at different stages of SN-CH communication as dictated by the protocols. In simulations, we incorporated processing delays for SNs and CHs based on measurements from the sea trials detailed in the previous section.

All authentication requests are transmitted as broadcast and may be processed by multiple CHs if an SN is within the overlapping coverage areas of these CHs. As the CHs form a blockchain network, authentication decisions are reached through consensus. Typically, only the CH nearest to the node responds, unless packet loss or collisions occur. If multiple CHs receive a request, they reach a consensus on which one will authenticate and respond. This consensus is established via high-speed, over-the-air radio communication. Once authenticated, an SN can transmit data in encrypted form using a session key shared with the corresponding CH. This session key has a validity of 10 minutes. Upon expiration, the SN is prompted to initiate a new authentication procedure.

To effectively simulate the blockchain component within the DESERT Underwater simulator, we employed the Redis data platform [85], chosen for its robust capability to support atomic operations on complex data types. As CHs process authentication requests, they inject the data into Redis, which serves as a simulated ledger. This setup mimics the decentralized and consensus-driven nature of blockchain technology, ensuring efficient and precise handling of authentication data without external locking mechanisms, thus maintaining the integrity of the multi-threaded DESERT environment.

In our analysis, we examined overhead introduced by the implemented protocols. Specifically, we measured the total energy consumption of the nodes during transmission and reception activities associated with authentication and re-authentication processes. Furthermore, we measured the protocol latency by recording the average time required for a node to authenticate/re-authenticate with CHs, which provided insights into the efficiency of the protocols under various network setups.

Figure 8 illustrates the average energy consumption for the BEKMP protocol compared to the protocol proposed in [22]. As anticipated, energy consumption increases with network size due to more nodes involved in communication and a higher likelihood of packet collisions. Notably, Figure 8 demonstrates that the BEKMP protocol is significantly more efficient, with its total energy consumption at least 92.8% lower than that of the protocol referenced in [22]. This substantial reduction is attributed to BEKMP's use of smaller payloads and a decreased packet count. Additionally, unlike the protocol [22], which employs a streamlined re-authentication process during cross-cluster mobility, the BEKMP protocol eliminates the need for re authentication altogether. If a node transitions from one CH's coverage area to another, the new CH can immediately decrypt the data since all CHs maintain critical security information in a shared ledger.

Figure 9 illustrates the time required for network nodes to successfully establish a secure channel with a CH, including

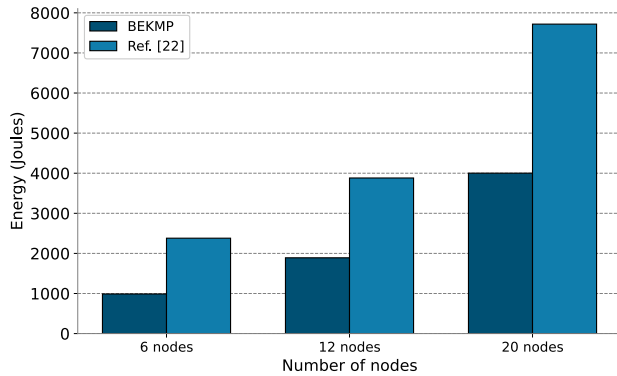


FIGURE 8. SN energy consumption for different UASN topologies.

propagation, transmission and processing delays for all exchanged packets. It is important to note that processing time is negligible compared to these delays, given the underwater sound speed of approximately 1500 m/s and the low bit-rate of the acoustic modems. The results clearly indicate that the BEKMP protocol, with its smaller communication overhead, establishes secure channels significantly faster than the protocol proposed in [22]. Specifically, across all the tested topologies, the BEKMP protocol is, on average, at least 73.2% faster than the protocol from [22].

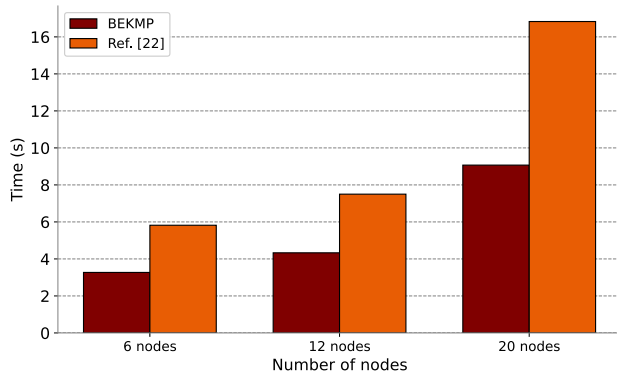


FIGURE 9. Protocol latency for different UASN topologies.

VI. CONCLUSION

In this paper, we proposed a blockchain-assisted key management protocol designed to ensure secure and low-latency authentication in underwater acoustic sensor networks. This protocol utilizes the inherent security features of blockchain technology to eliminate the need for re-authentication of sensor nodes during cross-cluster transitions, thereby fortifying the network against both external and internal threats. Central to our proposed security framework is the use of lightweight ECQV implicit certificates and the HOMQV protocol for authenticated key exchanges, making it particularly well-suited for resource-constrained devices. Our solution has undergone validation through both simulations and empirical experiments, demonstrating its robustness, efficiency, and scalability.

In our future research, we plan to extend our experiments to encompass broader network configurations. We aim to leverage the heterogeneous robotic swarm framework

described in [83], which incorporates aPad surface vehicles and non-anchored underwater sensor nodes, to conduct deeper and more comprehensive experimental analyses. This setup could be made more comprehensive by involving AUVs or simulating their presence using more accessible ROVs. Furthermore, we intend to integrate advanced Artificial Intelligence (AI) techniques to enhance the detection and mitigation of internal threats within these networks. Another promising avenue for future investigation involves adapting our solutions to UASN scenarios that demand long-term secrecy. Given the limitations of the HOMQV protocol in fulfilling these requirements, we plan to assess the feasibility of alternative security methods that can offer the necessary level of protection for resource-constrained underwater devices. It is crucial to evaluate how these security measures affect overall authentication efficiency, especially focusing on the impacts related to introduced delays and increased energy consumption.

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